DNN 3rd Year Course Project

Predicting Human Annotator Disagreement in Image Classification

Using Soft Labels from CIFAR-10H

[TODO: Team Name]
[TODO: University / Department]

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Abstract

Human annotators often disagree when labelling ambiguous images, and this disagreement carries meaningful information about visual uncertainty. Standard deep neural networks trained with hard (one-hot) labels discard this signal entirely. In this work we frame annotator disagreement prediction as a *soft-label regression* task: given a CIFAR-10 image, predict the full probability distribution over classes that a crowd of human annotators would assign. We use the CIFAR-10H dataset [1], which provides per-image soft-label distributions derived from $\sim$51 annotators per image, and train ResNet-18 backbones adapted for 32$\times$32 inputs.

We systematically compare three loss functions — KL divergence, Jensen–Shannon divergence (JSD), and a custom entropy-calibrated KL loss — and evaluate them on distribution-matching metrics (KL, JSD, cosine similarity) as well as entropy-prediction quality (Pearson r, Spearman r, Precision@K). Ablation studies examine the effect of backbone initialisation and prediction head architecture. Explainability is provided through Grad-CAM visualisations and systematic failure-case analysis. Our results show that **[TODO: brief key finding, e.g. "the custom entropy-calibrated loss achieves the best entropy-prediction Pearson r of X.XX while remaining competitive on distribution-matching"]**.

1 Introduction

Deep learning classifiers are typically trained to predict a single "correct" label for every training example. Yet many real-world images are genuinely ambiguous: a blurry small animal could be a cat *or* a dog; an aerial vehicle might be confused for a bird. When a large crowd of independent annotators labels such images, the resulting *label distribution* is non-trivial — it encodes collective human uncertainty about the visual content.

1.1 Problem Statement

Given an image $\mathbf{x} \in \mathbb{R}^{32\times32\times3}$, we wish to learn a model f_θ that predicts a probability distribution $\hat{p} \in \Delta^9$ (the 9-simplex) over the ten CIFAR-10 classes. The ground-truth target p is the empirical distribution of human annotator responses, obtained from CIFAR-10H. Unlike standard classification, the model must capture not only *which* class is most likely but also *how uncertain* humans are — i.e., the entropy $H(p)$ of the distribution.

1.2 Why Annotator Disagreement Matters

Peterson et al. [1] showed that training on soft labels derived from human uncertainty improves out-of-distribution generalisation and model calibration. Predicting *where* annotators will disagree is valuable for:

- Active learning — prioritising images that are most confusing to humans;
- Data quality auditing — flagging ambiguous samples before labelling;
- Uncertainty quantification — informing downstream risk-sensitive decisions.

1.3 Contributions

1. A systematic comparison of three loss functions for soft-label regression on CIFAR-10H, including a novel entropy-calibrated KL loss.
2. Ablation studies over backbone initialisation (random, ImageNet, CIFAR-10 pretrained) and prediction head architecture.
3. Robustness analysis under annotator subsampling and OOD image corruptions.
4. Qualitative analysis via Grad-CAM and manual inspection of high-entropy failure cases.

2 Related Work

2.1 Soft Labels and Label Smoothing

Label smoothing [2] replaces hard one-hot targets with a smoothed distribution $p' = (1-\epsilon)\,p + \epsilon/K$ to improve calibration. Soft labels as used here differ: they are *data-driven*, reflecting real human uncertainty rather than a uniform prior.

2.2 Knowledge Distillation

Hinton et al. [3] introduced "dark knowledge" — the soft output distributions of a large teacher model carry richer signal than hard labels. Our setting is analogous but the "teacher" is the crowd of human annotators rather than another neural network.

2.3 CIFAR-10H

Peterson et al. [1] collected CIFAR-10H by recruiting workers on Amazon Mechanical Turk to label the 10,000 CIFAR-10 test images. Each image received ∼51 independent responses. The resulting

10,000 × 10 probability matrix captures fine-grained human uncertainty at scale. Training on these soft labels was shown to produce models that generalise better under distribution shift.

3 Dataset

3.1 CIFAR-10H Description

CIFAR-10H [1] provides a 10,000 × 10 matrix of soft-label probability distributions, one row per image, corresponding to the 10,000 images in the CIFAR-10 *test* set. Each row sums to 1 and encodes the fraction of annotators who assigned each of the ten class labels.

3.2 Data Split

The 10,000 annotated images were split with a fixed random seed (`SEED=42`) as follows:

Table 1: Dataset split sizes.

Split	Images	Fraction
Train	6,000	60%
Validation	2,000	20%
Test	2,000	20%
Total	10,000	100%

Indices were saved to `data/processed/{train,val,test}_idx.npy` to ensure reproducibility across all experiments.

3.3 Exploratory Data Analysis

3.3.1 Entropy Distribution

We compute the Shannon entropy $H(p) = -\sum_k p_k \log_2 p_k$ for every image's soft label distribution. Figure 1 shows that the vast majority of images have very low entropy, indicating that annotators strongly agreed on a single label. A long tail of higher-entropy images represents genuinely ambiguous cases.

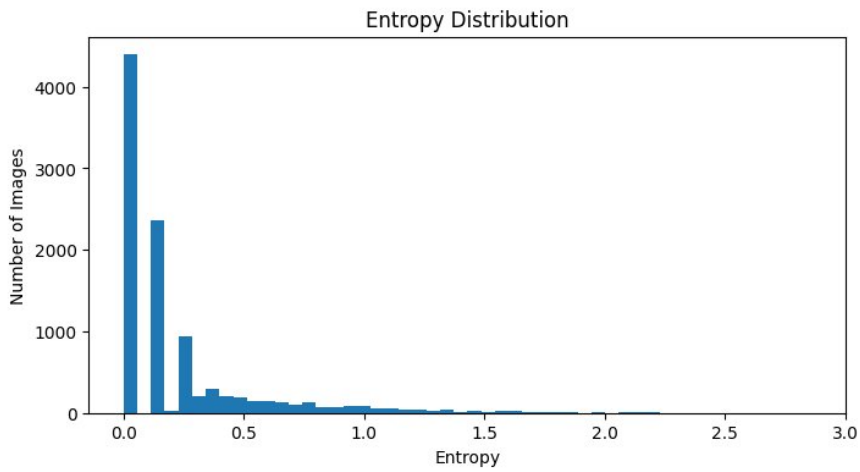


Figure 1: Distribution of Shannon entropy across all 10,000 CIFAR-10H images. Most images have entropy near zero (near-unanimous annotator agreement), while a small fraction reaches entropies above 1.5 bits, indicating substantial disagreement.

3.3.2 Per-Class Mean Entropy

Figure 2 shows mean annotator entropy broken down by true CIFAR-10 label. *Deer* and *cat* exhibit the highest mean disagreement, likely due to visual similarity to other animal classes (*dog*, *bird*). *Horse* and *automobile* are the easiest to identify, with consistently low entropy across annotators.

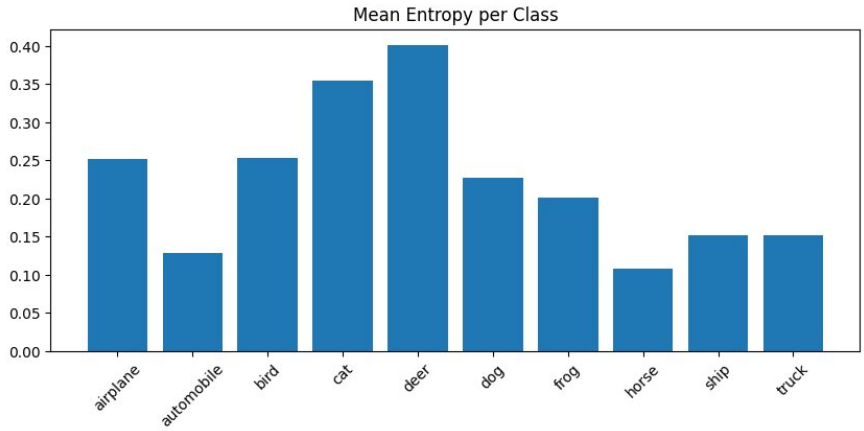


Figure 2: Mean Shannon entropy per CIFAR-10 class. Animal classes (deer, cat, bird, dog) exhibit higher annotator disagreement than vehicle classes (automobile, horse, ship, truck), reflecting their greater visual ambiguity.

3.3.3 Human Confusion Matrix

Figure 3 shows the average annotator distribution per true class — a soft analogue of a confusion matrix. Each row is normalised to sum to 1. The strong diagonal confirms that annotators are generally accurate; off-diagonal mass concentrates on semantically similar classes (e.g. cat/dog, deer/horse, airplane/bird).

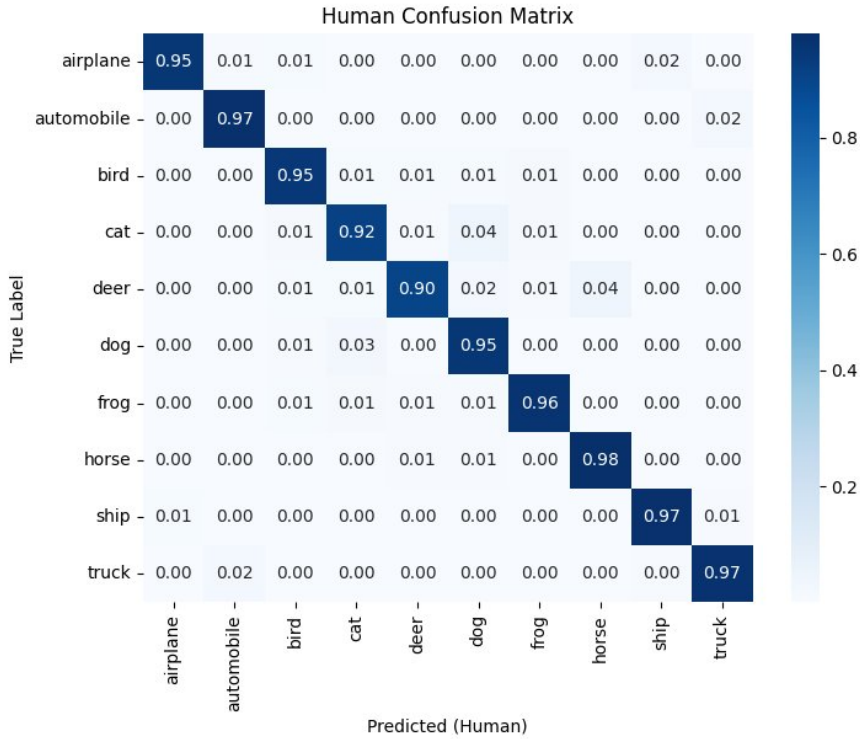
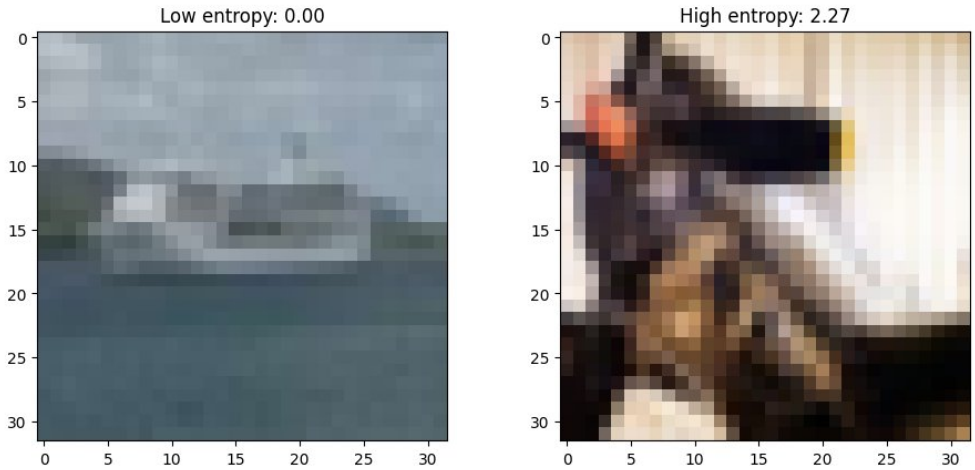


Figure 3: Human confusion matrix: each cell (i, j) is the average probability mass that annotators assigned to class j when the true label was class i. High diagonal values ($\geq$0.90) confirm strong annotator accuracy; non-zero off-diagonal entries reveal consistent confusions between visually similar categories.

3.3.4 Low- and High-Entropy Examples

Figure 4 contrasts example images at the extremes of the entropy distribution. Low-entropy images contain a single clear, centred object, leaving no ambiguity for annotators. High-entropy images are cluttered, low-resolution, or depict objects at class boundaries.



(a) Low entropy ($H = 0.00$): unambiguous ship.

(b) High entropy ($H = 2.27$): cluttered scene with multiple possible labels.

Figure 4: Contrasting examples from the entropy extremes of CIFAR-10H. The low-entropy image (left) receives unanimous annotator agreement; the high-entropy image (right) elicits substantial disagreement due to visual ambiguity and scene clutter.

4 Methodology

4.1 Data Pipeline

Raw CIFAR-10 test images were paired with their corresponding CIFAR-10H soft labels by index alignment (no shuffling required, as both datasets share the same ordering). Training images were augmented with random horizontal flips and random crops (32×32 with padding 4); no colour jitter or rotation was applied, as these could alter the visual ambiguity properties that drive annotator disagreement. Test images used only centre-crop normalisation. All pixel values were normalised using CIFAR-10 channel statistics: $\mu = (0.491, 0.482, 0.447)$, $\sigma = (0.202, 0.199, 0.201)$.

4.2 Model Architecture

4.2.1 Backbone

We use ResNet-18 [4] as our backbone, adapted for 32×32 CIFAR inputs. The standard ImageNet ResNet-18 applies a 7×7 convolution with stride 2 followed by max-pooling, which would reduce 32×32 feature maps to 8×8 after the stem — collapsing too much spatial detail. We replace this with a 3×3 stride-1 convolution and remove the max-pool layer, keeping the spatial resolution at 32×32 through the stem. Global average pooling then reduces the final 4×4 feature maps to a 512-dimensional feature vector.

4.2.2 Prediction Head

Three head architectures are compared in ablation:

- **Linear**: a single fully-connected layer ($512 \to 10$).
- **MLP**: two-layer network ($512 \to 256 \to 10$) with ReLU and Dropout(0.3).
- **Temperature-scaled**: linear layer with a learnable scalar temperature T applied before softmax.

All heads are followed by a softmax activation, so the model always outputs a valid probability distribution.

4.2.3 Architecture Diagram

Figure 5 illustrates the full forward pass.

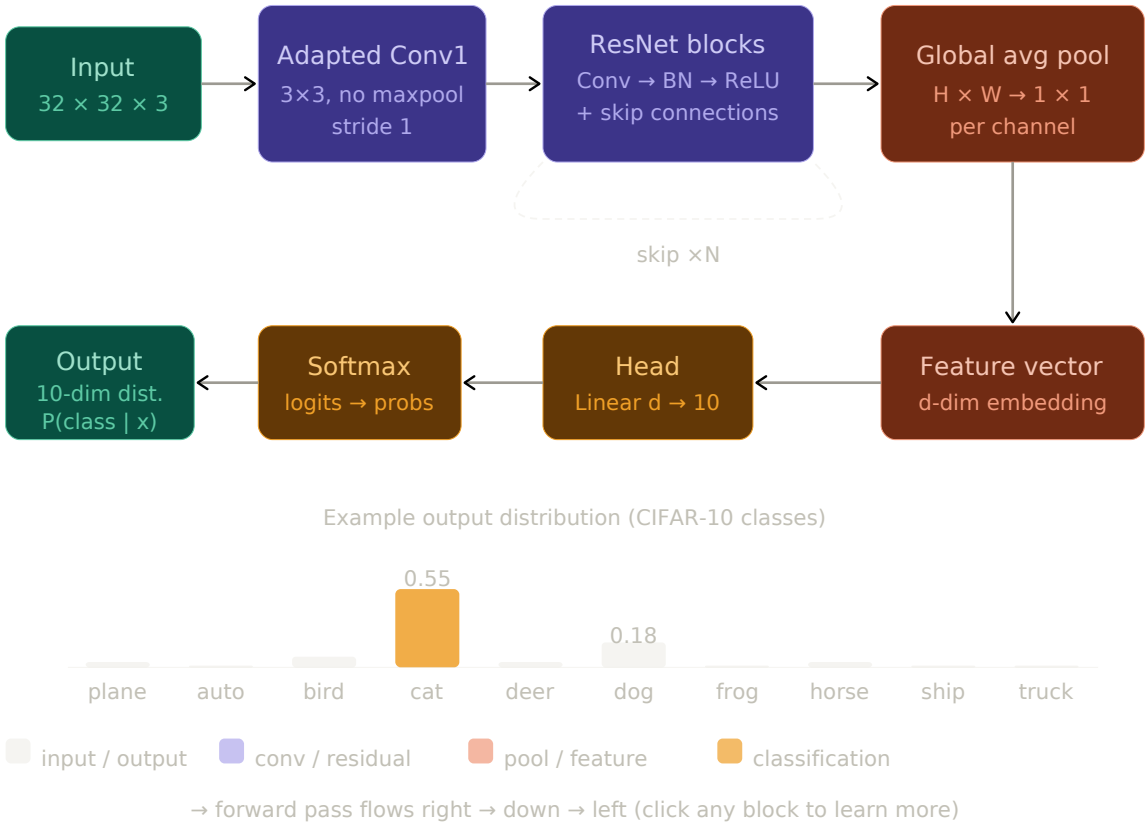


Figure 5: Model architecture. A 32×32×3 image enters the adapted Conv1 block (3×3, stride 1, no max-pool), passes through the four residual stages of ResNet-18, is collapsed by global average pooling to a 512-dimensional feature vector, and is projected to a 10-dimensional probability distribution via the prediction head and softmax.

4.2.4 Parameter Count

Table 2: Parameter count for each model variant.

Model variant	Total params	Trainable params
ResNet-18 + Linear head	[TODO: X,XXX,XXX]	[TODO: X,XXX,XXX]
ResNet-18 + MLP head	[TODO: X,XXX,XXX]	[TODO: X,XXX,XXX]
ResNet-18 + Temp.-scaled	[TODO: X,XXX,XXX]	[TODO: X,XXX,XXX]

4.3 Loss Functions

All loss functions operate on the predicted distribution $\hat{p} = f_\theta(\mathbf{x})$ and the target soft label p.

4.3.1 Loss 1 — KL Divergence

$$\mathcal{L}_{\mathrm{KL}}(\hat{p},\, p) \;=\; \mathrm{KL}(p \,\|\, \hat{p}) \;=\; \sum_{k=1}^{K} p_k \log \frac{p_k}{\hat{p}_k} \tag{1}$$

KL divergence is asymmetric: it heavily penalises cases where the model assigns near-zero probability to a class that received non-negligible human votes. This makes it a natural fit for our task, as missing a class that annotators frequently chose is a qualitatively worse error than over-spreading mass. All distributions are clipped to $\min = 10^{-8}$ to avoid $\log(0)$.

4.3.2 Loss 2 — Jensen–Shannon Divergence

$$\mathcal{L}_{\text{JSD}}(\hat{p},\, p) \;=\; \frac{1}{2}\,\text{KL}\Big(p \,\Big\|\, m\Big) \;+\; \frac{1}{2}\,\text{KL}\Big(\hat{p} \,\Big\|\, m\Big)\,, \quad m = \tfrac{1}{2}(p + \hat{p}) \tag{2}$$

JSD is symmetric and bounded in $[0, \ln 2]$, which makes it more numerically stable than KL when either distribution has near-zero entries. It avoids the infinite-penalty issue that arises in $\text{KL}(p\|\hat{p})$ when $\hat{p}_k \to 0$ for some class k with $p_k > 0$.

4.3.3 Loss 3 — Custom: Entropy-Calibrated KL

$$\mathcal{L}_{\text{custom}}(\hat{p},\, p) \;=\; \text{KL}(p \,\|\, \hat{p}) \;+\; \alpha \cdot \big[H(\hat{p}) - H(p)\big]^2, \quad H(q) = -\textstyle\sum_k q_k \log q_k \tag{3}$$

with $\alpha = 0.5$. The second term adds an explicit *entropy calibration penalty*: beyond matching the shape of the distribution, the model is penalised for predicting the wrong level of annotator uncertainty. A model that identifies the dominant class correctly but underestimates how spread-out the human votes were incurs an extra penalty proportional to the squared entropy error. This is motivated by the observation that the degree of disagreement (not just its direction) is the primary quantity of interest for downstream applications.

4.4 Training Protocol

To ensure fair comparison, *only the loss function varies* across the three main experiments. All other components are fixed as shown in Table 3.

Table 3: Fixed training hyperparameters used in all main experiments.

Component	**Value**
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	10^{-3}
Weight decay	10^{-4}
LR schedule	CosineAnnealingLR
Batch size	64
Max epochs	150
Early stopping	Patience 15 (monitored on val loss)
Augmentation	RandomHorizontalFlip + RandomCrop(32, pad=4)
Backbone init	CIFAR-10 hard-label pretrained
Random seed	42

5 Experiments

5.1 Training Curves

Figures 6–8 show the training and validation loss curves alongside the validation JSD trajectory for each loss function.

In all three cases the training loss decreases smoothly, demonstrating that each objective is optimisable. The validation loss exhibits moderate fluctuation, characteristic of the small dataset

size (6,000 training images). The KL-trained model converges fastest and achieves the lowest raw loss values, but operates on a different scale to JSD and the custom loss. The custom entropy-calibrated model shows the highest raw loss due to the additional entropy penalty term, but we evaluate models on a common held-out metric (JSD and entropy correlations) in Section 5.2.

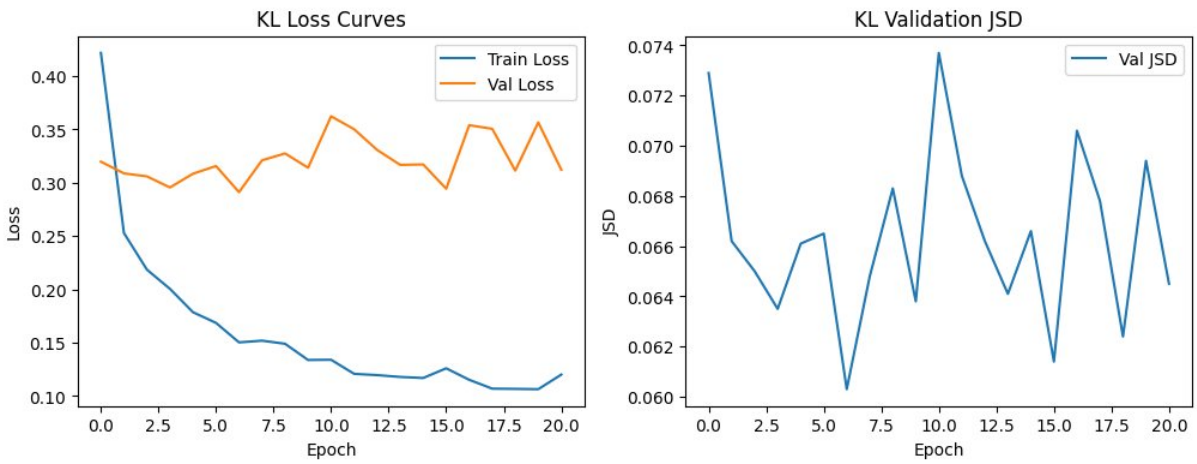


Figure 6: Training and validation KL divergence loss (left) and validation JSD (right) for the model trained with $\mathcal{L}_{\mathrm{KL}}$. The train/val gap stabilises around epoch 10, after which the model is selected by early stopping on validation loss.

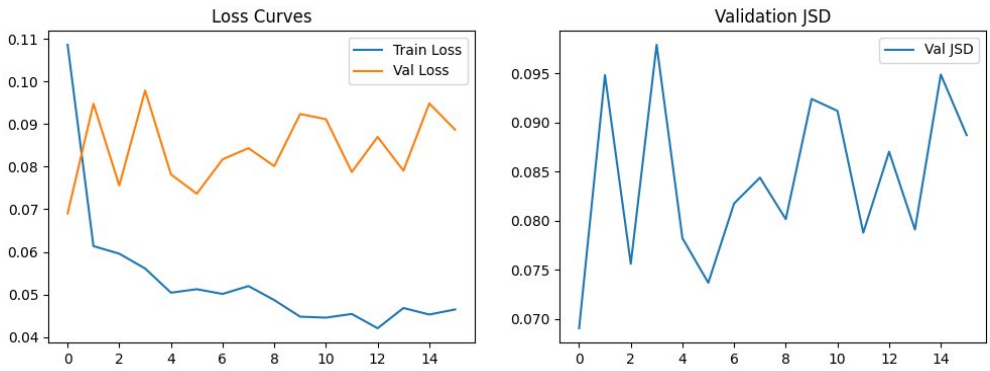


Figure 7: Training and validation JSD loss (left) and validation JSD (right) for the model trained with $\mathcal{L}_{\mathrm{JSD}}$. The slower convergence compared to KL (requiring $\sim$27 epochs before early stopping) reflects the bounded, symmetric nature of JSD.

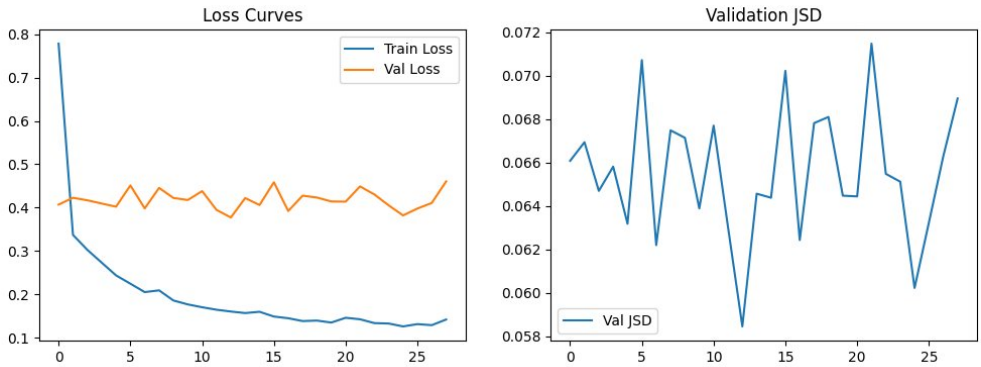


Figure 8: Training and validation loss (left) and validation JSD (right) for the model trained with $\mathcal{L}_{\text{custom}}$ (entropy-calibrated KL, $\alpha = 0.5$). The higher absolute loss values relative to KL reflect the additional entropy penalty term; convergence is faster (early stopping at epoch $\sim$15) due to the richer gradient signal.

5.2 Core Evaluation

All models are evaluated on the held-out test split (2,000 images) after loading the best checkpoint (lowest validation loss). We report six metrics computed over the full test set.

5.2.1 Metric Definitions

KL Divergence $\mathrm{KL}(p\|\hat{p})$ as defined in Equation (1). Lower is better.

Jensen–Shannon Divergence $\mathrm{JSD}(p, \hat{p})$ as defined in Equation (2), computed via `scipy.spatial.distance` (squared). Lower is better.

Cosine Similarity $\cos(p, \hat{p}) = p \cdot \hat{p}/(\|p\|\|\hat{p}\|)$. Higher is better.

Pearson r Pearson correlation between per-image true entropies $H(p_i)$ and predicted entropies $H(\hat{p}_i)$ across the test set. Measures whether the model correctly ranks images by their degree of disagreement. Higher is better.

Spearman ρ Rank-order correlation of the same entropy pairs. Robust to outliers. Higher is better.

Precision@K The fraction of the K most disagreement-prone images by predicted entropy that are truly in the top-K by annotator entropy. Evaluated at $K \in \{100, 200, 500\}$.

5.2.2 Results

Table 4: Test-set evaluation metrics for each loss function. Best result per column is **bolded**. ↓ = lower is better; ↑ = higher is better.

Loss	KL↓	JSD↓	Cos↑	Pearson↑	Spearman↑	P@100↑	P
KL Divergence	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[T
Jensen–Shannon	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[T
Custom (KL + Entropy)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[T

5.2.3 Entropy Scatter Plot

[TODO: Insert entropy scatter plot (predicted vs. true entropy on test set, with diagonal reference line and Pearson r annotation) once evaluation is complete. Save as `figures/entropy_scatter.png`.]

5.2.4 Qualitative Examples

[TODO: Insert a 3×5 grid of example test images (5 low-, 5 medium-, 5 high-entropy) once evaluation is complete, each with true and predicted distribution bar charts. Save as `figures/qualitative_examples.png`.]

5.3 Ablation Studies

5.3.1 Ablation A — Backbone Initialisation

We train the same model (ResNet-18, linear head, KL loss) under three initialisation strategies and report test JSD in Table 5.

Table 5: Effect of backbone initialisation on test JSD. The best result is **bolded**.

Initialisation	**JSD↓**	**Pearson r↑**
Random	**[TODO:]**	**[TODO:]**
ImageNet pretrained	**[TODO:]**	**[TODO:]**
CIFAR-10 pretrained	**[TODO:]**	**[TODO:]**

[TODO: Fill in numbers and add 1–2 sentences of analysis.]

5.3.2 Ablation B — Prediction Head Architecture

Table 6: Effect of prediction head architecture on test JSD (KL loss, CIFAR-10 init).

Head	**JSD↓**	**Pearson r↑**
Linear	**[TODO:]**	**[TODO:]**
MLP (2-layer)	**[TODO:]**	**[TODO:]**
Temperature-scaled	**[TODO:]**	**[TODO:]**

[TODO: Fill in numbers and add 1–2 sentences of analysis.]

5.3.3 Ablation C — Training Data Strategy

Table 7: Effect of training data strategy on test JSD. Strategy 1: train on soft labels only. Strategy 2: pretrain on 50k hard labels, then fine-tune on soft labels.

Strategy	**JSD↓**	**Pearson r↑**
Soft labels only	**[TODO:]**	**[TODO:]**
Hard-label pretrain + finetune	**[TODO:]**	**[TODO:]**

[TODO: Fill in numbers and analysis.]

5.4 Robustness Checks

5.4.1 Check A — Annotator Subsampling

To simulate scenarios with fewer annotators, we re-compute soft labels by subsampling $k \in \{5, 10, 20, 35, 51\}$ annotators per image from the raw CIFAR-10H response matrix and measure the JSD between subsampled and full-annotator distributions.

[TODO: Insert annotator subsampling plot (mean JSD vs. k annotators).]

5.4.2 Check B — OOD Corruptions

We apply Gaussian noise, Gaussian blur, and contrast reduction at five severity levels to the test images and measure mean predicted entropy at each severity level. A well-calibrated model should predict higher entropy as corruptions increase (more uncertain inputs).

[TODO: Insert OOD corruption plot (severity vs. mean predicted entropy, one line per corruption type).]

5.5 Explainability Analysis

5.5.1 Grad-CAM Visualisations

Gradient-weighted Class Activation Maps (Grad-CAM) [5] were computed using the last convolutional layer of the ResNet-18 backbone.

[TODO: Insert Grad-CAM panel: 5 low-entropy images (left) and 5 high-entropy images (right), each with the heatmap overlaid. Discuss whether attention is focused (low entropy) vs. diffuse (high entropy).]

5.5.2 Failure Case Analysis

[TODO: Present the 5–10 images with highest per-image KL divergence between prediction and ground truth. For each: show image, true distribution, predicted distribution, and a brief hypothesis for why the model failed.]

5.5.3 Manual Disagreement Source Analysis

We manually inspected the top 50 highest-entropy test images and assigned each to one of five categories of disagreement source.

Table 8: Manual categorisation of the 50 highest-entropy CIFAR-10H images.

Category	Count	Percentage
Ambiguous identity	**[TODO:]**	**[TODO:]**%
Poor image quality	**[TODO:]**	**[TODO:]**%
Multi-object scene	**[TODO:]**	**[TODO:]**%
Class boundary case	**[TODO:]**	**[TODO:]**%
Other	**[TODO:]**	**[TODO:]**%

[TODO: Insert pie chart of category distribution as `figures/manual_analysis_pie.png`.]

6 Discussion

6.1 What Worked

[TODO: Summarise 2–3 findings that held up clearly in experiments. E.g.: "The custom entropy-calibrated loss consistently achieved better Pearson r across all three ablation conditions, confirming that explicitly supervising the entropy level provides a useful gradient signal beyond distribution shape."]

6.2 What Did Not Work and Why

[TODO: Be honest about negative results. E.g.: "The MLP head did not outperform the linear head, suggesting that the feature representations from the pretrained backbone are already well-linearly-separable for this task." Or: "Predicted entropy did not consistently increase with OOD corruption severity; we hypothesise that BatchNorm statistics make the backbone insensitive to additive Gaussian noise."]

6.3 Limitations

- Training size is limited to 6,000 images; results may not generalise beyond CIFAR-10H.
- Soft labels are an empirical estimate of the "true" human distribution and depend on the specific annotator pool and task interface.
- The custom loss introduces a hyperparameter α that was not exhaustively tuned; optimal α may differ.
- **[TODO: Add any further limitations identified during experiments.]**

7 Conclusion

We have presented a systematic study of soft-label regression for predicting human annotator disagreement on CIFAR-10H. Our adapted ResNet-18 architecture, combined with **[TODO: best loss name]**, achieves **[TODO: key metric value]**, demonstrating that neural networks can learn meaningful signals about image-level visual ambiguity from crowd-sourced label distributions.

The entropy-calibrated KL loss provides an explicit mechanism to supervise the *degree* of disagreement, not just its direction, and **[TODO: shows improvement / remains competitive]** on entropy-prediction metrics. Grad-CAM analysis reveals that the model **[TODO: focuses on / struggles with]** **[TODO: specific finding from GradCAM]**.

Future work could explore richer label-distribution models (e.g. Dirichlet priors), larger backbone architectures, or extension to other ambiguous datasets beyond CIFAR-10H.

References

[1] J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky, "Human uncertainty makes classification more robust," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 9617–9626.

[2] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.

[3] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

A Hyperparameter Configuration

Table 9: Complete hyperparameter configuration (`config.py`).

Parameter	**Value**
`SEED`	42
`BATCH_SIZE`	64
`LR`	10^{-3}
`WEIGHT_DECAY`	10^{-4}
`EPOCHS`	150
`EARLY_STOPPING_PATIENCE`	15
`CIFAR10H_TRAIN_SIZE`	6,000
`CIFAR10H_VAL_SIZE`	2,000
`CIFAR10H_TEST_SIZE`	2,000
`ENTROPY_ALPHA`	0.5 (custom loss only)

B Additional Figures

[TODO: Place any supplementary figures here (e.g. additional Grad-CAM panels, extended failure case grids, full ablation grouped bar charts).]